

Solution to Ramanujan Challenge Problem 2.1: A Polynomial Continued Fraction for $6/(3 - \pi)$

Xiang Huang
University of Illinois Springfield
xhuan5@uis.edu

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Abstract

We prove that the polynomial continued fraction of Problem 2.1 equals $6/(3 - \pi)$. The solution has two elementary ingredients and one classical input. The elementary ingredients are (i) a polynomial index shift $a_n = -\alpha(n + 1)$, $b_n = \beta(n)$ identifying the challenge's coefficients with those of Entry 5.3.22 of Cohen's dictionary of polynomial continued fractions, and (ii) a sign-flip lemma: negating every partial denominator of a continued fraction, while keeping the partial numerators, negates every convergent. Because (ii) is proved *at the level of convergents*, no question of tail convergence arises. The classical input is the value of Cohen's Entry 5.3.22 itself, namely $\pi = 3 + 6/T$ where T is the corresponding continued fraction; the challenge's fraction is exactly the sign-flip of T , hence equals $-T = 6/(3 - \pi)$. The argument is machine-checked in Lean 4, with Entry 5.3.22 appearing as an explicit hypothesis.

1 The problem

Let

$$a_n = -220n^3 - 484n^2 - 301n - 42, \tag{1}$$

$$b_n = 4n^2(2n + 1)^2(5n - 4)(5n + 6). \tag{2}$$

The challenge asks for a proof that

$$a_0 + \frac{b_1}{a_1 + \frac{b_2}{a_2 + \frac{b_3}{a_3 + \ddots}}} = \frac{6}{3 - \pi}. \tag{3}$$

Throughout, for coefficient sequences $(c_n)_{n \geq 0}$ and $(d_n)_{n \geq 1}$ we write P_n, Q_n for the convergents of $c_0 + d_1/(c_1 + d_2/(c_2 + \cdots))$, defined by the classical recursions

$$\begin{aligned} P_{-1} &= 1, & P_0 &= c_0, & P_n &= c_n P_{n-1} + d_n P_{n-2}, \\ Q_{-1} &= 0, & Q_0 &= 1, & Q_n &= c_n Q_{n-1} + d_n Q_{n-2}, \end{aligned} \tag{4}$$

and the value of the continued fraction means $\lim_n P_n/Q_n$ when it exists.

2 The classical input

Theorem 2.1 (Cohen, Entry 5.3.22). *Let*

$$\alpha(n) = 220n^3 - 176n^2 - 7n + 5, \quad \beta(n) = 4n^2(2n+1)^2(5n-4)(5n+6). \quad (5)$$

Then

$$\pi = 3 + \frac{6}{\alpha(1) + \frac{\beta(1)}{\alpha(2) + \frac{\beta(2)}{\alpha(3) + \ddots}}} = 3 + \frac{6}{42 + \frac{396}{1047 + \frac{38400}{4340 + \ddots}}}. \quad (6)$$

This is Entry 5.3.22 of Cohen's *Continued Fractions of Polynomial Type: Theory and Encyclopedic Dictionary* [2], where it is recorded in the machine-readable form

```
[() -> Pi, [3, 220*n^3-176*n^2-7*n+5],
  [6, 4*n^2*(2*n+1)^2*(5*n-4)*(5*n+6)]]
```

together with the displayed expansion whose first partial quotients are 42, 396, 1047, 38400, 4340, $\dots$; the same fraction appears as entry 1.2.18 of the earlier database [1]. Cohen obtains it from the hypergeometric seed for $\pi = 6 \arcsin(1/2)$ by a Bauer–Muir/Apéry acceleration followed by an even contraction and a polynomial equivalence transformation. This is the one step of the present solution that we cite rather than reprove; everything else below is proved here.

We have checked (6) numerically to 128 decimal digits (§7), and verified the coefficients (5) against the source entry, including its displayed values 42, 396, 1047, 38400, 4340.

3 The index shift

Lemma 3.1. *For all $n \geq 0$,*

$$a_n = -\alpha(n+1), \quad b_n = \beta(n).$$

Proof. The second identity holds because b_n and $\beta(n)$ are the same expression. For the first,

$$\begin{aligned} \alpha(n+1) &= 220(n+1)^3 - 176(n+1)^2 - 7(n+1) + 5 \\ &= (220n^3 + 660n^2 + 660n + 220) - (176n^2 + 352n + 176) - 7n - 7 + 5 \\ &= 220n^3 + 484n^2 + 301n + 42, \end{aligned}$$

so $-\alpha(n+1) = a_n$. □

Consequently, writing

$$c_n := \alpha(n+1), \quad d_n := \beta(n),$$

the continued fraction $T := c_0 + d_1/(c_1 + d_2/(c_2 + \dots))$ is exactly the tail appearing in (6), and the challenge's fraction (3) is the one with partial denominators $-c_n$ and the *same* partial numerators d_n .

4 The sign-flip lemma

Lemma 4.1. *Let $(c_n), (d_n)$ be arbitrary, with convergents P_n, Q_n as in (4), and let $\tilde{P}_n, \tilde{Q}_n$ be the convergents of the continued fraction with partial denominators $-c_n$ and partial numerators d_n . Then for all $n \geq -1$,*

$$\tilde{P}_n = (-1)^{n+1} P_n, \quad \tilde{Q}_n = (-1)^n Q_n.$$

Consequently $\tilde{P}_n/\tilde{Q}_n = -P_n/Q_n$ for every n .

Proof. Induction on n . For the base cases, $\tilde{P}_{-1} = 1 = (-1)^0 P_{-1}$, $\tilde{P}_0 = -c_0 = (-1)^1 P_0$, $\tilde{Q}_{-1} = 0 = (-1)^{-1} Q_{-1}$ and $\tilde{Q}_0 = 1 = (-1)^0 Q_0$. For the step, assuming the claim at $n-1$ and $n-2$,

$$\tilde{P}_n = (-c_n)\tilde{P}_{n-1} + d_n\tilde{P}_{n-2} = (-c_n)(-1)^n P_{n-1} + d_n(-1)^{n-1} P_{n-2} = (-1)^{n+1}(c_n P_{n-1} + d_n P_{n-2}) = (-1)^{n+1} P_n,$$

and identically

$$\tilde{Q}_n = (-c_n)(-1)^{n-1} Q_{n-1} + d_n(-1)^{n-2} Q_{n-2} = (-1)^n (c_n Q_{n-1} + d_n Q_{n-2}) = (-1)^n Q_n.$$

Finally $\tilde{P}_n/\tilde{Q}_n = (-1)^{n+1} P_n / ((-1)^n Q_n) = -P_n/Q_n$. $\square$

We emphasise that Lemma 4.1 is an identity for each fixed n . It involves no hypothesis about convergence, and in particular no manipulation of continued-fraction tails, which is where sign arguments of this kind usually become delicate. It follows at once that the two continued fractions converge or diverge together, and that when they converge their values are negatives.

5 The main theorem

Theorem 5.1. *With a_n, b_n as in (1)–(2),*

$$a_0 + \frac{b_1}{a_1 + \frac{b_2}{a_2 + \ddots}} = \frac{6}{3 - \pi}.$$

Proof. By Theorem 2.1, the continued fraction T with partial denominators $c_n = \alpha(n+1)$ and partial numerators $d_n = \beta(n)$ converges, and $\pi = 3 + 6/T$; since $\pi \neq 3$ this gives

$$T = \frac{6}{\pi - 3}.$$

By Lemma 3.1, the challenge's continued fraction has partial denominators $a_n = -c_n$ and partial numerators $b_n = d_n$, so it is precisely the sign-flip of T in the sense of Lemma 4.1. By that lemma its n -th convergent equals $-P_n/Q_n$ for every n , hence it converges, with value

$$-T = -\frac{6}{\pi - 3} = \frac{6}{3 - \pi}. \quad \square$$

Remark 5.2. Numerically $6/(3 - \pi) = -42.3750798355862746187580309154\dots$, and the convergence is geometric at the golden-ratio rate: Cohen records convergence type φ^{-10} per term, i.e. $10 \log_{10} \varphi = 2.0899\dots$ decimal digits per term, which is what we observe (§7). In particular the challenge's fraction converges at exactly the same rate as Cohen's, as Lemma 4.1 forces.

6 Formal verification

The accompanying Lean 4 development (`lean/`, toolchain `leanprover/lean4:v4.30.0`) builds with **0 errors and 0 sorry** and formalises everything above except Theorem 2.1:

<code>shift_a, shift_b</code>	Lemma 3.1
<code>alphaC_one... b21_one</code>	the displayed values 42, 396, 1047, 38400
<code>cfP, cfQ</code>	the convergent recursions (4)
<code>cfP_neg, cfQ_neg</code>	Lemma 4.1
<code>cf_neg_convergent</code>	$\tilde{P}_n/\tilde{Q}_n = -P_n/Q_n$, for every n
<code>challenge_convergent_eq</code>	the challenge is the sign-flip of Cohen’s tail
<code>problem21_pcf_value</code>	Theorem 5.1, given Theorem 2.1

All of these depend only on the three standard axioms `propext`, `Classical.choice`, `Quot.sound`. There is no `native_decide`. Theorem 2.1 enters as an explicit hypothesis:

```
theorem problem21_pcf_value
  (hCohen : Tendsto (fun k => cfP cohenC cohenD k / cfQ cohenC cohenD k)
    atTop (nhds (6 / (Real.pi - 3)))) :
  Tendsto (fun k => cfP a21 b21 k / cfQ a21 b21 k)
    atTop (nhds (6 / (3 - Real.pi)))
```

so that the reliance on Cohen’s entry is legible in the statement itself rather than hidden in an axiom or a `sorry`.

7 Numerical verification

The script `verify.py` reimplements the same convergent recursion (4) used in the Lean file and checks, in exact rational arithmetic where possible:

1. the index-shift identities of Lemma 3.1 for $0 \leq n < 200$;
2. the coefficient values against Cohen’s displayed expansion: $\alpha(1) = 42$, $\beta(1) = 396$, $\alpha(2) = 1047$, $\beta(2) = 38400$, $\alpha(3) = 4340$, and $a_0 = -42$, $a_1 = -1047$, $b_1 = 396$;
3. the sign-flip identity $\tilde{P}_n = (-1)^{n+1}P_n$, $\tilde{Q}_n = (-1)^nQ_n$ *exactly*, for $0 \leq n \leq 41$;
4. convergence of (6) to π : the error at $n = 61$ is 5.4×10^{-129} ;
5. convergence of the challenge’s fraction to $6/(3 - \pi)$: the error at $n = 61$ is 1.6×10^{-126} , matching the predicted 2.09 digits per term.

References

- [1] H. Cohen, *A Database of Continued Fractions of Polynomial Type*, arXiv:2409.06086, 2024.
- [2] H. Cohen, *Continued Fractions of Polynomial Type: Theory and Encyclopedic Dictionary*, arXiv:2607.06581, 2026. Entry 5.3.22.
- [3] W. B. Jones and W. J. Thron, *Continued Fractions: Analytic Theory and Applications*, Encyclopedia of Mathematics and its Applications 11, Addison–Wesley, 1980.
- [4] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.